HW # 12

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$x - y - 2z = 0$

$x = y + 2z$

$$x = \begin{bmatrix} y+2z \\ y \\ z \end{bmatrix} \Rightarrow x = \underbrace{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}}_{X_1} y + \underbrace{\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}}_{X_2} z$$

$$V_1 = X_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$V_2 = X_2 - \frac{X_2 \cdot X_1}{X_1 \cdot X_1} \cdot X_1$$

$$V_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} - \frac{2}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Orthogonal basis: $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

$$N_1 = \frac{V_1}{\|V_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$V_2 = \frac{V_2}{\|W_2\|} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

Orthonormal basis: $\left\{ \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}, \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix} \right\}$